Experiment 9

Photoelectric Effect

Objective	**Materials Used**
Examine the Photoelectric Effect	Photoelectric Effect Emitters, Receiver, and Power Supplies

Theory

In 1905 Albert Einstein gave a simple explanation of Lenard's discoveries using Planck's theory. The new 'quantum'-based model predicted that higher frequency light would produce higher energy emitted electrons (photoelectrons), independent of intensity, while increased intensity would only increase the number of electrons emitted (or photoelectric current). Einstein assumed that the light shining on the emitter material could be thought of as 'quanta' of energy (called photons) with the amount of energy equal to hc/λ with λ as the wavelength. In the photoelectric effect, one 'quantum' of energy is absorbed by one electron. If the electron is below the surface of the emitter material, some of the absorbed energy is lost as the electron moves towards the surface. This is usually called the 'work function' (ϕ). If the 'quantum' is more than the 'work function', then the electron is emitted with a certain amount of kinetic energy. Einstein applied Planck's theory and explained the photoelectric effect in terms of the quantum model using his famous equation for which he received the Nobel prize in 1921:

$$E_{tot} = hf = KE_{max} + \phi$$

For our experiment, the maximum kinetic energy can be found measuring a stopping Voltage (V_{stop}) that just barely makes the current zero. The device we use is calibrated by wavelength instead of frequency, there for the above equation for our experiment become:

$$E_{tot} = hc/\lambda = eV_{stop} + \phi$$

Procedure

1. Calibration: Before beginning the experiment, familiarize yourself with the equipment. The mercury lamp needs some time to heat up. Turn it on 10 minutes before collecting data.

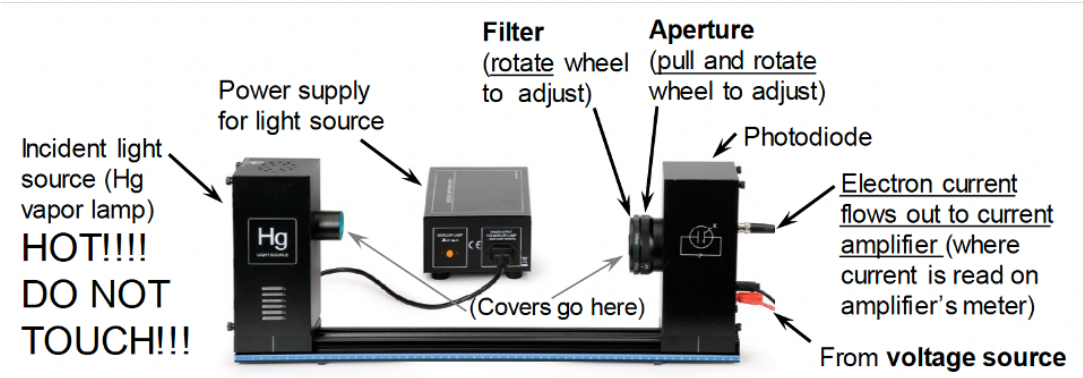

Figure 9.1: Reference Michelson Interferometer

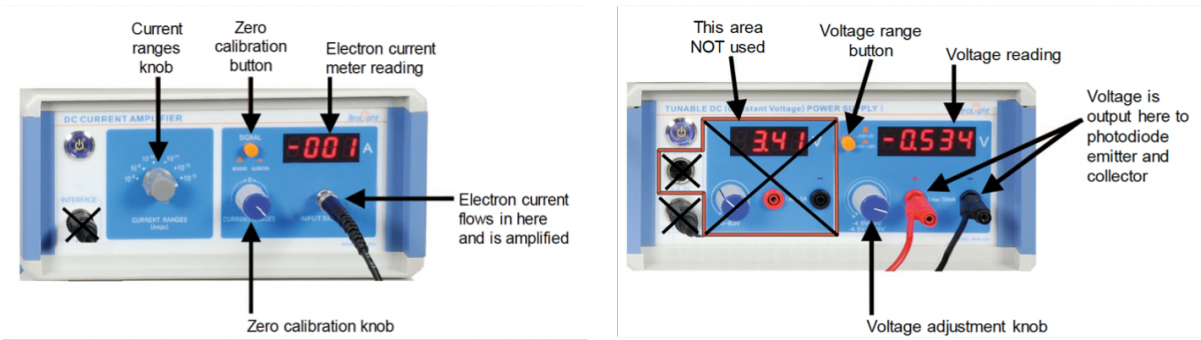

Figure 9.2: Ammeter and Stopping Voltage

This experiment measures currents on the order of 10^{-11} amps. Due to this, follow the calibration instructions below *before starting each experiment.*

(a) Confirm that the mercury light has its cover on. Confirm that the filter has its cover off.

(b) Set the current range knob to the appropriate setting (10^{-11} for procedure 1 and 10^{-13} for procedures 2 and 3).

(c) Push **IN** the measure/calibration button.

(d) Turn the **current adjust** knob until the current just barely reaches zero. Note that when you go below zero, the sign will change. Use this to get as close to zero as possible.

(e) Push **OUT** the measure/calibration button.

(f) The device is now ready to measure currents.

2. Procedure 1: Current vs Voltage

   (a) Perform the calibration steps outlined above.

   (b) Change the aperture to 2 mm (slight pull then rotate the wheel).

   (c) Change the filter to 436 nm (rotate the wheel).

   (d) Set the voltage range button to **IN** position to output a range of -4.50 V to 30.0 V . Set the voltage to -4.00 V.

   (e) Remove the cover of the mercury lamp to begin the experiment.

   (f) Measure the current of the system when the voltage is set to -4.00 V and -2.00 V. Be careful with your units.

   (g) Measure the stopping voltage for the system.

   (h) Measure the current for voltages between 0.00 V and 30.0 V while changing the voltage by 2 V in each step. Record this in a data table with your previous voltage measurements.

   (i) Replace the mercury lamp cover.

3. Procedure 2: Stopping Voltage vs Intensity

   (a) Perform the calibration steps above. For this experiment, set the current range to 10^{-13}

   (b) Set the aperture to 2 mm and the filter to 436 nm.

   (c) Set the voltage range button to the **OUT** position to output a range of -4.5 V to 0 V

   (d) Set the voltage to -4.00 V and remove the mercury lamp cover.

   (e) The aperture determines how much light makes it into the photodiode. As the aperture increases, so does the intensity on the system. Measure the stopping voltage for each aperture setting.

   (f) Replace the mercury lamp cover.

4. Procedure 3: Stopping Voltage vs Wavelength

   (a) Perform the calibration steps above. For this experiment, set the current range to 10^{-13}

   (b) Set the aperture to 8 mm and the wavelength to 365 nm.

   (c) Set the voltage range button to the **OUT** position to output a range of -4.5 V to 0 V

   (d) Set the voltage to -4.00 V and remove the mercury lamp cover.

   (e) Measure the stopping voltage for each wavelength listed on the filter. The stopping voltage is defined as the limit when the current goes to zero. Note that these values will be negative, but the stopping voltage should be measured as a positive value. Record these values in a data table with appropriate units.

   (f) Replace the mercury lamp cover.

Analysis

1. Create a plot of your data as current vs voltage. In an ideal case, we expect this to be a linear relationship. Is your graph linear? What key features can you identify from the graph?

2. Compare the stopping voltages against intensity. Does the stopping voltage change? Explain why this occurs.

3. Create a linear plot from your data of wavelength and stopping voltage. From this plot, measure the work func¡on and Planck's constant. Be mindful of your units. Compare your measured value of Planck's constant to the accepted value.